

MOHD ANAS

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EDUCATION

B.Tech, Information Technology

IIIT Vadodara, Gujarat, India

Expected Graduation: Aug 2027

7.46 / 10.0 GPA

Coursework: Data Structures, Algorithms, Operating Systems, Object-Oriented design, Distributed Systems, Backend Systems, Machine Learning, Artificial Intelligence

PROFESSIONAL EXPERIENCE

Aarogya Kaya | SDE Intern | Remote

July 2025 - Jan 2026

- Architected Go backend using CQRS, PostgreSQL, and Supabase RPCs for efficient and centralized data access.
- Designed a multi-tiered backend architecture using in-memory cache, Redis, and database, reducing average read latency from **420ms to 38ms (91%)** and maintaining **p95 latency below 250ms**.
- Translated evolving product requirements into backend designs across 15+ modules, contributing through the **full SDLC**.
- Applied **SOLID principles and design patterns**, reducing code duplication by **30%**.
- Engineered fault-tolerant asynchronous synchronization pipelines with retries and idempotent processing, supporting 500+ daily events while maintaining 99.9% data consistency.

Open Source Contributor | Remote

July 2025 - Present

- **Contributed to Next.js (Vercel)** by migrating the patch-next utility from CommonJS to TypeScript ESM, eliminating repeated dependency reinstalls and reducing contributor setup overhead.
- Contributed 22 merged Pull Requests and opened 12 Issues across open-source projects, incorporating maintainer feedback to fix bugs, implement features, and improve developer tooling.

PROJECTS

a3.store | Distributed Key-value Store | Python, EKS, AWS, Kubernetes, Prometheus, Security

Live | GitHub

- Developed a leaderless **Distributed key-value store** implementing replication, failure detection, synchronization, gossip-based membership, and anti-entropy repair across a 3-node cluster.
- Deployed on AWS EKS using Kubernetes StatefulSets and persistent volumes, enabling **rolling upgrades, pod recovery** and **self-healing behavior**.
- Instrumented **production-grade observability** using Prometheus, Grafana, and Loki for real-time monitoring.

Shop-Intelligence | Real-Time Customer Behavior Analytics | Python, Computer Vision, Queue, Pytest

Live | GitHub

- Built a distributed real-time **computer vision pipeline** for four synchronized CCTV streams using concurrent workers and message queues, **processing 15 FPS** while reducing frame-processing latency to **50 ms**.
- Designed an **event-driven** analytics engine and global visitor registry that generated real-time backend events for customer behavior dashboards and retail insights.
- **Optimized multi-camera processing** using **multiprocessing** and **multithreading** to maximize throughput and ensure scalable real-time analytics.

TECHNICAL SKILLS

Languages: Go, Python, C/C++, Javascript, SQL

Frontend: React.js, React Flow, Next.js, TypeScript

Backend & System: FastAPI, Node.js, gRPC, REST APIs, Kafka, Redis

Cloud & Infrastructure: AWS, EKS, Kubernetes, Docker, Terraform, Linux, IAM, EC2, VPC, Git, GitHub

AI & LLMs: LangChain, HuggingFace, TensorFlow, PyTorch, Neural Network, DeepLearning, NLP, Generative AI

ACHIEVEMENT

- Ranked **365th** among **15,804** participants in CodeChef **START192D**.
- Ranked **3,026th** among **17,000+** participants in Codeforces Round 1110 (Div. 1 + Div. 2)

LEADERSHIP & ACTIVITIES

- Organized an intra-college hackathon and mentored **25+ teams** as a **Technical Committee Core Member**.
- Led recruitment, mentorship, and event initiatives for **100+ students** as **Joint Secretary of DOT Club**.